

7. Project Demonstration Phase

7.1 Introduction

The demonstration phase presents the completed ComicCraft application and shows its major functionalities in a real-world usage scenario. It begins with registration and login and continues through story input, script generation, character definition, panel artwork generation, speech bubble editing, and export.

7.2 Objectives of Demonstration

- To demonstrate the overall functionality of ComicCraft.
- To show the user authentication system.
- To demonstrate AI script generation from a story idea.
- To demonstrate character management and consistent panel artwork.
- To demonstrate speech bubble editing, layout, and export.
- To display the gallery of saved comics.
- To show the interaction between frontend, backend, database, and Gemini models.
- To present the final working application to the evaluator.

7.3 Demonstration Environment

The demonstration is performed in the configured development environment on a laptop with an internet connection, using VS Code to run the FastAPI server, a web browser for the frontend, the configured database, and a valid Google Gemini API key with access to text and image models.

7.4 Demonstration Workflow

Step	Feature	Demonstration
1	Registration & Login	Create an account and log in securely.
2	Story Input	Enter a story idea, genre, art style, and number of panels.
3	Script Generation	Generate the title, synopsis, and panel-by-panel script; edit a line of dialogue.
4	Characters	Review and adjust the character descriptions.
5	Panel Generation	Generate artwork for all panels and show consistent characters.
6	Regenerate	Regenerate one panel with a modified scene description.
7	Speech Bubbles	Add, move, and edit speech bubbles and captions.
8	Layout & Export	Preview the comic pages and export a PDF.
9	Gallery	Open previously saved comics.
10	Logout	End the session.

7.5 Sample Scenario

A school student enters the idea "a shy robot who learns to make friends at a space station school", selects a cartoon art style, and requests six panels. ComicCraft generates a title, a short synopsis, two characters, and a six-panel script. The student adjusts the robot's appearance description, generates the artwork, and notices that the fourth panel shows the wrong expression, so that panel is regenerated. The student then edits the speech bubbles, previews the pages, and exports the finished comic as a PDF to share with classmates.

7.6 Results and Observations

- The application meets the defined functional requirements.
- Gemini produces coherent, well-paced scripts in the requested format.
- Stored character descriptions improve consistency of characters across panels.
- Editable speech bubbles give clean, readable lettering.
- The interface is simple and easy to navigate.
- User data is protected through authentication.

7.7 Limitations

- Character consistency is improved but not perfect, and some panels may need regeneration.
- AI-generated images may contain minor visual errors.
- An internet connection and Gemini API access are required.
- Response time depends on API availability and network speed, and image generation is slower than text.
- Generated content must be used responsibly and may be subject to safety filters.

7.8 Future Enhancements

- Upload of reference images for characters.
- Support for multiple languages in dialogue.
- Voice narration and animated panels.
- Collaborative comic creation for multiple users.
- More art styles and page layout templates.
- Mobile application and print-ready export.

7.9 Conclusion

ComicCraft: AI Comic Story Creator using Gemini Models combines AI story scripting, consistent character management, panel image generation, editable speech bubbles, and comic export in a single application. The demonstration confirms that the system works as intended and gives users a convenient way to turn their story ideas into finished comics without drawing skills.